

# Cloud Computing Master Program

- Linux
- Aws
- DevOps

GIT

Jenkins

Ansible

Maven

Docker

# Cloud Computing Master Program

## Linux

- **Introduction to Linux**
    - Basics of Linux OS and its file system
    - Shell commands: ls, cp, mv, rm, etc.
    - Directory structure: /etc, /var, /home, etc.
    - Permissions and ownership: chmod, chown, chgrp
  - **User & Group Management**
    - Creating, modifying, and deleting users and groups
    - File and directory permissions
    - Managing sudoers, groups, and access control
  - **Package Management**
    - Installing and managing software using apt, yum, and rpm
    - Handling dependencies and package repositories
    - Installing from source
  - **System Monitoring and Performance**
    - System resource monitoring: top, htop, vmstat, iostat
    - Disk space management: df, du, mount, umount
    - Logs management: /var/log files, log rotation, journalctl
  - **Networking and Firewalls**
    - Basic networking concepts: IP addressing, DNS, routes
    - Configuring network interfaces: ifconfig, ip
    - Firewall setup with iptables and firewalld
  - **System Services and Processes**
    - Managing system services: systemctl, service, chkconfig
    - Monitoring and managing processes: ps, kill, bg, fg
    - Scheduling tasks: cron, at, and systemd timers
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# AWS

- **Overview of Cloud Computing**
  - Introduction to cloud computing models: IaaS, PaaS, SaaS
  - Benefits of AWS cloud: Scalability, security, cost-efficiency
  - Regions, Availability Zones (AZs), and Edge Locations
- **AWS Compute Services**
  - EC2 Instances: Launch, configure, and manage EC2 instances
  - Auto Scaling and Load Balancing: Scaling EC2 instances with Auto Scaling Groups, ELB
  - Lambda: Serverless architecture, creating and managing Lambda functions
- **AWS Storage Services**
  - S3: Buckets, objects, versioning, lifecycle policies, and permissions
  - EBS: Volumes, snapshots, mounting and data persistence
  - Glacier, EFS, and Storage Gateway
- **AWS Networking Services**
  - VPC: VPC setup, subnets, route tables, security groups, and NACLs
  - Elastic IPs and NAT Gateways
  - Direct Connect and VPN connections
- **AWS Security**
  - IAM: Users, groups, roles, policies, and MFA
  - AWS KMS: Key management and encryption
  - AWS Shield, WAF, and Security Best Practices
- **AWS Databases**
  - RDS: Relational databases, backup, and replication
  - DynamoDB: NoSQL database setup and management
  - ElastiCache, Redshift, Aurora
- **Monitoring and Logging in AWS**
  - CloudWatch: Monitoring, Alarming, and Logs
  - CloudTrail: Auditing and security logging

- AWS Trusted Advisor and Cost Explorer
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# DevOps

- **DevOps Concepts**
    - DevOps lifecycle: Development, Testing, Integration, Deployment, Monitoring
    - Continuous Integration (CI) & Continuous Delivery (CD)
    - Infrastructure as Code (IaC)
  - **Collaboration & Communication Tools**
    - Agile methodology and its relation to DevOps
    - Tools: Jira, Confluence, Slack for team communication
    - Version control practices and collaboration
  - **Automated Testing**
    - Unit testing, integration testing, and functional testing
    - Test Automation tools: Selenium, JUnit
  - **CI/CD Pipeline Design and Implementation**
    - Building CI pipelines for automated code testing, building, and deployment
    - CD pipeline strategies: Blue/Green, Canary Releases, and Feature Flags
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## GIT (Version Control)

- **Introduction to Version Control**
  - Basics of Git: Repositories, commits, branches, and merges
  - Git workflows: GitFlow, GitHub Flow
- **Basic Git Commands**
  - Cloning repositories: git clone
  - Adding, committing, and pushing changes: git add, git commit, git push
  - Branching and merging: git branch, git merge
- **Collaborative Workflow**
  - Pull requests and code reviews

- Handling merge conflicts
  - Git hooks: Pre-commit, post-commit, etc.
  - **Advanced Git**
    - Rebasing vs Merging
    - Git tags and releases
    - Stashing and resetting changes
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## **Jenkins (CI/CD Tool)**

- **Introduction to Jenkins**
    - Installing and configuring Jenkins
    - Jenkins architecture: Master-Slave, Executors
    - Jenkins UI: Overview of Jobs, Pipelines, Build History
  - **Building a Jenkins Pipeline**
    - Creating Freestyle Projects
    - Setting up pipeline as code with Jenkinsfile
    - Integrating with GitHub/GitLab for automatic trigger on commits
  - **Managing Jenkins Jobs**
    - Configuring build triggers: SCM polling, Webhooks, Scheduled triggers
    - Configuring build steps: Build tools (Maven, Gradle), running tests
  - **Managing Credentials and Security in Jenkins**
    - Jenkins Security: Users, roles, and permissions
    - Managing secrets using Jenkins credentials plugin
  - **Advanced Jenkins Pipelines**
    - Parallel stages, conditional builds, and build pipelines
    - Integrating Jenkins with other tools: Docker, Ansible, and Slack
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## **Ansible (Automation Tool)**

- **Introduction to Ansible**
  - Installing and configuring Ansible

- Understanding the architecture: Control node, managed nodes, inventory files
  - Playbooks, roles, and tasks
  - **Configuration Management with Ansible**
    - Writing Ansible Playbooks for automation
    - Defining tasks and handlers, using modules (file, user, apt, yum, etc.)
    - Ansible Vault for secure management of sensitive data
  - **Advanced Ansible Topics**
    - Managing inventories dynamically: Static and dynamic inventories
    - Working with Ansible Galaxy for reusable roles
    - Orchestrating multi-node configurations
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## **Maven (Build Automation Tool)**

- **Introduction to Maven**
    - Understanding Maven lifecycle: validate, compile, test, package, install, deploy
    - Basic Maven commands: mvn clean install, mvn compile
    - Project Object Model (POM): structure and configuration
  - **Building Projects with Maven**
    - Dependency management: Adding and managing dependencies in POM.xml
    - Building Java projects and packaging as JAR/WAR files
    - Configuring Maven plugins: Compiler, Surefire, Deploy
  - **Advanced Maven Configuration**
    - Multi-module projects
    - Profiles in Maven for different environments
    - Integration with CI/CD tools (Jenkins, GitLab CI)
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## **Docker (Containerization Tool)**

- **Introduction to Docker**
  - Understanding containers vs virtual machines

- Docker installation and setup
- Docker CLI: docker run, docker ps, docker images, docker exec
- **Working with Docker Containers**
  - Building Docker images using Dockerfile
  - Container lifecycle management: start, stop, restart
  - Managing Docker containers and images
- **Docker Networking and Volumes**
  - Container networking: Bridge, Host, Overlay networks
  - Using Docker volumes for persistent storage
  - Linking containers and managing multi-container applications
- **Docker Compose**
  - Introduction to Docker Compose for multi-container setups
  - Writing docker-compose.yml files for defining multi-container applications
  - Orchestrating services with Docker Compose
- **Docker Swarm and Kubernetes (Intro)**
  - Docker Swarm: Setting up clusters, services, and scaling
  - Introduction to Kubernetes for container orchestration